Epistemic Map Games (EMG):

Strategic Interaction without a Shared Objective Game

Fabrizio De Palma
Independent Researcher

Abstract

Recent work in applied game theory, particularly in cybersecurity and adversarial AI, highlights the difficulty of modeling strategic interaction when agents act through distorted or incomplete internal representations of their environment (e.g., Pawlick et al., 2019; Collins et al., 2024). In such settings, agents do not respond to a shared objective game, but to subjective representations of it.

We introduce **Epistemic Map Games (EMG)**, a minimal formalism in which each agent evaluates actions through an agent-specific epistemic map that generates subjective payoffs from an underlying game. Strategic equilibrium is defined directly over these subjective games, without assuming common knowledge of payoffs or representations.

We show that every finite EMG admits at least one equilibrium in mixed strategies. Classical Nash equilibrium is recovered as a limiting case when all epistemic maps coincide.

Keywords: Epistemic games; Epistemic map games; Subjective payoff games; Strategic misperception; Non-Bayesian uncertainty; Adversarial decision-making; Cybersecurity game theory; Cognitive asymmetry; Multi-agent systems; Adversarial AI.

---

1. Introduction

Classical game theory provides a coherent and mathematically elegant framework for strategic interaction, grounded in shared payoff structures, perfect rationality, and common knowledge. Over the past decades, numerous post-Nash extensions have relaxed some of these assumptions by introducing incomplete information, uncertainty, bounded rationality, or dynamic adaptation (e.g., Harsanyi, 1967). At the same time, classical epistemic results highlight the strength of the alignment assumptions required for consensus and equilibrium (Aumann, 1976).

However, recent work in applied game theory—particularly in cybersecurity, adversarial AI, and complex socio-technical systems—shows that these extensions still struggle to model operational environments in which agents act through **distorted, partial, or adversarially manipulated internal representations of the world** (e.g., Pawlick et al., 2019; Collins et al., 2024). In such settings, agents do not respond to a single objective game, but to **subjective internal maps** that shape how payoffs, risks, and opportunities are perceived.

Modern cyber interactions make this limitation explicit. Attackers, defenders, and autonomous agents routinely operate under incomplete situational awareness, deception, misinformation, and cognitive bias. Strategic behavior in these contexts is driven not by objective payoff matrices, but

by **agent-specific interpretations** of them. Classical and post-Nash models were not explicitly designed to capture this form of epistemic misalignment.

This paper introduces **Epistemic Map Games (EMG)**, a minimal generalization of normal-form games in which each agent evaluates strategies through an individual epistemic map that generates subjective payoffs from an underlying game. Strategic equilibrium is defined directly over these subjective games, without assuming common knowledge of payoffs or representations. In EMG, players need not optimize over the same mathematical object. This departs from the foundational assumption—shared since Nash—that strategic interaction presupposes a common game. From a formal standpoint, EMG questions whether the existence of a shared game object should be treated as a primitive assumption of strategic interaction, or rather as a special—and fragile—case of epistemic alignment.

The proposed framework preserves the core structure of Nash equilibrium while extending it to settings with heterogeneous internal representations. When all epistemic maps coincide, EMG reduces exactly to the classical Nash model. For finite games, we prove the existence of equilibrium in mixed strategies.

By shifting the modeling focus from objective payoff structures to subjective epistemic representations, EMG provides a unified and tractable framework for analyzing strategic interaction in cyber conflict, adversarial AI, and other domains characterized by persistent epistemic distortion.

---

2. Epistemic Map Games

2.1 Baseline Game (Nash, 1950)

Let

$$G = \langle N, (S_i)_{i \in N}, (u_i)_{i \in N} \rangle$$

be a finite normal-form game, where:

- $N = \{1, \dots, n\}$is the set of players,
- $S_i$is the finite set of pure strategies available to player i,
- u_i: $\prod_{j \in N} S_j \to \mathbb{R}$is the objective payoff function of player i.

This baseline game is not assumed to be directly accessible to the players. It serves only as a reference structure from which subjective evaluations are generated. The baseline game serves only as a structural reference, not as a cognitive object shared by the players.

2.2 Epistemic Maps and Subjective Payoffs

For each player $i \in N$, we define an **epistemic map**

$$\phi_i : \mathbb{R} \to \mathbb{R},$$

which transforms objective payoffs into subjective evaluations.

Given a strategy profile $s \in \prod_{j \in N} S_j$, player $i$evaluates outcomes according to the **subjective payoff function**

$$u_i^{\phi}(s) := \phi_i(u_i(s)).$$

No structural assumptions are imposed on the epistemic maps beyond measurability. No stronger regularity assumptions are required for the equilibrium existence result; continuity may be imposed without loss of generality in applied settings. In particular, maps may differ across players and need not preserve order, linearity, or affine structure.

2.3 Epistemic Map Game

An **Epistemic Map Game (EMG)** is defined as the tuple

$$G^{\phi} = \langle N, (S_i)_{i \in N}, (u_i^{\phi})_{i \in N} \rangle,$$

where subjective payoffs are generated via player-specific epistemic maps as defined above.

Players choose strategies so as to maximize their subjective payoffs u_i^{ϕ}, not the objective payoffs u_i.

2.4 Mixed Strategies and Equilibrium

Let $\Delta(S_i)$denote the set of mixed strategies over S_i.
For a mixed strategy profile $\sigma = (\sigma_1, \dots, \sigma_n)$, subjective expected payoffs are given by:

$$U_i^{\phi}(\sigma) := \mathbb{E}_{s \sim \sigma}[u_i^{\phi}(s)].$$

A mixed strategy profile $\sigma^*$is an **Epistemic Map Equilibrium (EME)** if, for every player $i \in N$,

$$U_i^{\phi}(\sigma^*) \geq U_i^{\phi}(\sigma_i, \sigma_{-i}^*) \forall \sigma_i \in \Delta(S_i).$$

2.5 Relation to Nash Equilibrium

If $\phi_i$is the identity map for every $i \in N$, then $u_i^{\phi} = u_i$and the EMG reduces exactly to the original normal-form game G. In this case, EME coincides with the classical Nash equilibrium.

Thus, Nash games arise as a special case of EMG under perfect epistemic alignment.

2.6 Existence of Equilibrium

Since $G^{\phi}$is a finite normal-form game with real-valued payoffs, standard fixed-point arguments imply the existence of at least one mixed-strategy EME.

The existence result is deliberately unsurprising: the conceptual novelty lies not in equilibrium existence, but in the object over which equilibrium is defined.

A formal proof is provided in **Appendix A**.

---

3. Toy Examples

This section illustrates how Epistemic Map Games generate equilibrium outcomes that differ from classical Nash predictions, even in simple finite games. The purpose of these examples is not empirical validation, but structural illustration of epistemic misalignment effects.
All examples are deliberately minimal and are intended to highlight structural effects rather than empirical realism.

3.1 Prisoner's Dilemma with Epistemic Distortion

Consider the standard Prisoner's Dilemma with payoff matrix:

	C	**D**
C	(3,3)	(0,5)
D	(5,0)	(1,1)

In the classical game, mutual defection $(D, D)$is the unique Nash equilibrium.

Assume now that both players apply epistemic maps ϕ_1, ϕ_2that amplify cooperative outcomes relative to defection, e.g.:

$$\phi_i(3) > \phi_i(1), \phi_i(0) \ll \phi_i(1).$$

Under such maps, the subjective payoff ranking may invert the dominance structure, making cooperation a best response to cooperation.
As a result, (C, C)may constitute an Epistemic Map Equilibrium even though it is not a Nash equilibrium of the baseline game.

This outcome is not due to communication, repetition, or enforcement, but solely to epistemic distortion of payoffs.

3.2 Poker-like Bluffing Game

Consider a simplified two-player zero-sum game where one player may bluff.
In classical models, bluffing equilibria rely on mixed strategies and common knowledge of payoff structure.

In an EMG, a player's epistemic map may overestimate losses associated with being called, or underestimate gains from successful bluffing.
The opponent's epistemic map may, in turn, distort the perceived likelihood of bluffing.

The resulting equilibrium differs from the classical mixed-strategy equilibrium, even though strategy sets and objective payoffs are unchanged.

This illustrates how EMG captures strategic misalignment without introducing types, beliefs, or probabilistic uncertainty.

3.3 Cooperation under Asymmetric Perception

Consider a coordination game where one player's epistemic map coincides with the objective payoff function, while the other's does not.

Even in the absence of private information or signaling, equilibrium strategies may differ from Nash predictions, as one player optimizes against a perceived game that does not exist objectively.

This asymmetry alone is sufficient to alter equilibrium selection.

3.4 Single-Agent Case (Degenerate EMG)

EMG naturally extends to the case $n = 1$.

Let an agent choose among actions with objective payoffs $u(a)$, but evaluate them through an epistemic map ϕ.
Decision-making occurs entirely within the subjective payoff space.

This reduces EMG to a minimal theory of decision-making under epistemic distortion, encompassing perceptual bias, misvaluation, and bounded rationality without invoking stochastic beliefs.

3.5 Discussion

Across all examples, equilibrium deviations from classical predictions arise solely from epistemic misalignment, not from information asymmetry, belief hierarchies, or dynamic effects.

These examples show that EMG isolates a structural source of strategic divergence that is orthogonal to existing post-Nash refinements.

---

4. Conclusion

This paper introduced a minimal formal framework in which strategic interaction is modeled over **agent-specific epistemic maps** rather than a single, commonly shared representation of the game. In adversarial domains, assuming a shared objective game is often the exception rather than the rule. Within this setting, we defined the **Epistemic Map Equilibrium (EME)** as a natural generalization of Nash equilibrium when players optimize with respect to their own subjective payoff structures.

The framework is intentionally minimal.
No assumptions are made about the origin of epistemic distortions, their probabilistic structure, or their dynamics.
When all epistemic maps coincide with the objective payoff structure, EME collapses to the standard Nash equilibrium.
When at least one agent operates on a distorted epistemic map, the equilibrium concept departs from Nash while preserving its core fixed-point logic.

The contribution of this work is not a new behavioral model or a refinement of rationality assumptions, but a **structural shift** in how strategic interaction is represented: players need not optimize over the same game object.
This perspective is particularly aligned with adversarial and security-oriented settings, where agents routinely act under divergent, incomplete, or distorted representations of the environment.

EME should be read as a foundational step.
Extensions to dynamic settings, multi-agent environments, observer effects, and domain-specific applications are natural directions for future work, but are deliberately left outside the scope of this paper.

Appendix A. Existence of EME (finite case)

For completeness, we restate the existence result in a style parallel to the classical proof for Nash equilibrium.

Let

$$G^R = (\{A, D\}, S_A, S_D, v_A, v_D)$$

be a finite epistemic-map game. Define the mixed-strategy spaces

$$X_A = \Delta(S_A), X_D = \Delta(S_D), X = X_A \times X_D.$$

For each $i \in \{A, D\}$, the subjective expected payoff

$$V_i : X \to \mathbb{R}$$

is multilinear and continuous. Define the best-response correspondence $B_i : X \to 2^{X_i}$by

$$B_i(\sigma_A, \sigma_D) = \arg \max_{\sigma_i' \in X_i} V_i(\sigma_i', \sigma_{-i}).$$

Then $B = B_A \times B_D : X \to 2^X$is non-empty-valued, convex-valued and has closed graph. By Kakutani's fixed-point theorem, there exists $\sigma^* \in X$such that $\sigma^* \in B(\sigma^*)$, which is an EME.

Note that the proof does not depend on any particular form of the epistemic maps R_i, only on the finiteness of strategy sets and boundedness of payoffs.

Appendix B.

Immediate Application Domains (illustrative only)

This appendix lists application domains in which the EMG framework is immediately applicable.
The list is non-exhaustive and intentionally non-technical: its purpose is to show the breadth of interactions naturally captured by epistemic-map-based games, without committing to specific implementations or empirical validations.

The unifying feature of all cases below is the interaction between agents (or between an agent and its environment) operating on **subjective epistemic maps** rather than on a shared, objective representation of the world.

Cybersecurity and Information Warfare

- Cyber attack–defense interactions under incomplete, distorted, or deceptive situational awareness
- Advanced Persistent Threats (APT) with asymmetric reconnaissance and belief formation
- Deception-based defense (honeypots, decoys, moving target defense)
- Cognitive hacking and misinformation attacks
- Multi-stage intrusion modeling (e.g., Cyber Kill Chain) with divergent attacker/defender perceptions
- Security operations under partial or conflicting telemetry
- Adversarial machine learning and model extraction attacks
- Red teaming and blue teaming with non-aligned threat models

---

Military and Strategic Interactions

- Deterrence under misperception or strategic ambiguity
- Hybrid warfare and gray-zone conflicts
- Signaling and bluffing in escalation scenarios
- Asymmetric conflicts with unequal situational awareness
- Command-and-control decisions under fragmented intelligence
- Psychological operations and influence campaigns
- Strategic surprise and miscalculation dynamics

Artificial Intelligence and Autonomous Systems

- Multi-agent AI systems with heterogeneous world models
- Human–AI interaction with epistemic misalignment
- Autonomous decision-making under partial observability
- Reinforcement learning agents acting on learned world models
- Adversarial environments for autonomous agents
- Coordination failures due to divergent internal representations

---

Economics and Finance

- Market interactions under heterogeneous beliefs
- Speculative behavior driven by distorted expectations
- Strategic trading with asymmetric perception of fundamentals
- Asset pricing under epistemic disagreement
- Negotiation and bargaining with misaligned valuations
- Behavioral finance phenomena beyond probabilistic noise

Political Science and International Relations

- Electoral competition with asymmetric voter perception
- Strategic communication and propaganda
- Coalition formation under belief divergence
- Diplomatic negotiation under strategic misinterpretation
- Public opinion dynamics as an epistemic field

Social Systems and Networks

- Social influence and opinion formation
- Polarization driven by divergent epistemic maps
- Rumor spreading and belief cascades
- Collective behavior under informational fragmentation
- Platform-mediated interactions with algorithmic distortion

Biology and Natural Systems

- Virus–host interactions modeled as agent–environment games
- Immune response under incomplete pathogen information
- Evolutionary dynamics with limited environmental sensing
- Adaptive behavior in simple organisms

Human Decision-Making and Cognition

- Decision-making under perceptual distortion
- Bias-driven strategic interaction
- Learning under misrepresentation
- Human–machine cooperative decision systems

Remark.
In all listed domains, uncertainty is not merely probabilistic but epistemic and structural: agents act on representations that may be incomplete, distorted, or strategically influenced. EMG provides a unified formal framework to model such interactions without requiring a shared or objective world representation.

References

Aumann, R. J. (1976). Agreeing to disagree. *The Annals of Statistics*, 4(6), 1236–1239.

Collins, A., Ferguson, R., & Bolstad, R. (2024). Constructing shared understanding of complex interdisciplinary problems: Epistemic games in interdisciplinary teamwork. *Journal of the Learning Sciences*, 33(2), 405–442.

Harsanyi, J. C. (1967). Games with incomplete information played by "Bayesian" players, I–III. *Management Science*, 14(3), 159–182.

Nash, J. F. (1950). Equilibrium points in n-person games. *Proceedings of the National Academy of Sciences of the United States of America*, 36(1), 48–49.

Pawlick, J., Zhu, Q., & Rieger, C. (2019). A survey of deception in cybersecurity. *Proceedings of the IEEE Symposium on Security and Privacy*, 549–563.